

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1786

Chatom 138 kV
June 18, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1786. The Interconnection Customer proposes a 70.4 MW solar photovoltaic facility interconnecting at the Chatom 138 kV substation in the Virginia area. The request was received on June 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 627 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,168,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

MIDDLETOWN JCT - MOUNT PLEASANT SUBSTATION 138kV: 144.6% loading (220 MVA) following the outage of the FT BENTON - MOUNT PLEASANT 138kV L; DFAX = 5.5%.

The MOUNT PLEASANT - FT BENTON 138kV is overloaded to 133.7% of its 229 MVA emergency rating for the loss of the FORT BENTON - MIDDLETOWN JUNCTION 138kV L. The project contributes 7.9% DFAX (5.5 MW) to this overload.

CHATOM - ALVIN 138kV: 132.4% loading (397 MVA) following the outage of the PAGE - PURDY 138kV Line; DFAX = 30.1%.

For the loss of MOUNT PLEASANT - FIREBANZ 138kV line, the MOUNT PLEASANT - SOUTH JORDAN SUBSTATION 138kV Line loads to 130.2% of its 232 MVA rating. Project DFAX: 6.4%.

For the loss of FT BENTON - MOUNT PLEASANT 138kV L, the FORT BENTON - MIDDLETOWN JCT 138kV line loads to 129.6% of its 219 MVA rating. Project DFAX: 7.8%.

The NORTHPOINT - MITCHELL GARDENS 138kV Line is overloaded to 112.7% of its 513 MVA emergency rating for the loss of the PAGE - PURDY 138kV. The project contributes 20.3% DFAX (14.3 MW) to this overload.

For the loss of PAGE - PURDY 138kV line, the NORTHPOINT - S BELLEVILLE 138kV Line loads to 112.3% of its 532 MVA rating. Project DFAX: 20.3%.

The NUCLEAR FUELS SUBSTATION - ALVIN 138kV is overloaded to 109.2% of its 475 MVA emergency rating for the loss of the PAGE - PURDY 138kV line. The project contributes 30.1% DFAX (21.2 MW) to this overload.

The TES FILER CITY STA - JETPORT 138kV L is overloaded to 109.0% of its 183 MVA emergency rating for the loss of the TES FILER CITY STA - BRIDGE RIV SUBSTATION 138kV. The project contributes 5.7% DFAX (4.0 MW) to this overload.

The ABBOTSFORD - BRIDGE RIVER 138kV is overloaded to 108.2% of its 386 MVA emergency rating for the loss of the BROOKLYN CITY N PLANT - MIDET 138kV Line. The project contributes 6.9% DFAX (4.9 MW) to this overload.

For the loss of SPRINGFIELD POWER LLC - EDINBURG 138kV, the PLANT CITY - RED TAIL SUBSTATION 138kV loads to 106.7% of its 278 MVA rating. Project DFAX: 9.7%.

ST. MARTINS - WADSWORTH 138kV: 106.6% loading (314 MVA) following the outage of the ST LOUIS - NUCLEAR FUELS 138kV; DFAX = 20.3%.

The TEMPLETON - RED TAIL 138kV is overloaded to 106.1% of its 288 MVA emergency rating for the loss of the SPRINGFIELD POWER LLC - EDINBURG 138kV line. The project contributes 9.7% DFAX (6.9 MW) to this overload.

For the loss of SPRIGFIELD POWER LLC SUBSTATION - EDINBURG SUBSTATION 138kV line, the TEMPLETON - BUCKHORN SUBSTATION 138kV loads to 106.0% of its 289 MVA rating. Project DFAX: 9.7%.

ROSLYN - BUCKHORN 138kV: 105.9% loading (291 MVA) following the outage of the SPRINGFIELD POWER LLC - EDINBURG 138kV; DFAX = 9.7%.

The HOUSTON - ST LOUIS 138kV is overloaded to 105.2% of its 260 MVA emergency rating for the loss of the ST LOUIS - NUCLEAR FUELS 138kV. The project contributes 8.4% DFAX (5.9 MW) to this overload.

For the loss of BRIDGE RIVER - TES FILER CITY STATION 138kV line, the GLEN HEAD - DUNN 138kV loads to 103.2% of its 313 MVA rating. Project DFAX: 12.0%.

For the loss of BRIDGE RIVER - TES FILER CITY STA 138kV line, the DUNN - CEDARBURG SOUTH 138kV loads to 103.1% of its 316 MVA rating. Project DFAX: 12.0%.

ST LOUIS - NUCLEAR FUELS 138kV L: 103.1% loading (428 MVA) following the outage of the PURDY - PAGE 138kV; DFAX = 18.0%.

For the loss of AMERICAN - URBAN 345kV, the TES FILER CITY STATION - BRIDGE RIV 138kV loads to 103.0% of its 515 MVA rating. Project DFAX: 8.3%.

For the loss of PAGE - PURDY 138kV, the S BELLEVILLE - NICHOLS 138kV loads to 103.0% of its 589 MVA rating. Project DFAX: 20.3%.

The PLANT CITY - CEDARBURG S 138kV line is overloaded to 102.9% of its 287 MVA emergency rating for the loss of the SPRINGFIELD POWER LLC SUBSTATION - EDINBURG 138kV line. The project contributes 9.7% DFAX (6.9 MW) to this overload.

ST LOUIS - NICHOLS 138kV: 102.9% loading (595 MVA) following the outage of the PURDY - PAGE 138kV; DFAX = 20.3%.

BIDGE RIVER - MIDSET 138kV line: 102.8% loading (302 MVA) following the outage of the BRIDGE RIVER SUBSTATION - ABBOTSFORD 138kV; DFAX = 7.3%.

The PURDY - CHATOM 138kV L is overloaded to 102.5% of its 740 MVA emergency rating for the loss of the FRITO LAY SUBSTATION - DPC PRAIRIE U CHIEN 138kV. The project contributes 30.8% DFAX (21.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MIDDLETOWN JCT - MOUNT PLEASANT SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$9,591,000	6% (\$581,000)
FT BENTON SUBSTATION - MOUNT PLEASANT 138kV	Rebuild line with higher-capacity conductor	\$17,844,000	6% (\$1,078,000)
CHATOM - ALVIN 138kV	Rebuild line with higher-capacity conductor	\$3,361,000	6% (\$202,000)
S JORAN - MOUNT PLEASANT 138kV	Rebuild line with higher-capacity conductor	\$15,672,000	6% (\$1,014,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FORT BENTON - MIDDLETOWN JCT 138kV line	Rebuild line with higher-capacity conductor	\$14,387,000	6% (\$914,000)
MITCHELL GARDENS - NORTHPOINT 138kV	Rebuild line with higher-capacity conductor	\$7,506,000	5% (\$362,000)
NORTHPOINT - S BELLEVILLE 138kV Line	Rebuild line with higher-capacity conductor	\$5,706,000	5% (\$275,000)
NUCLEAR FUELS - ALVIN 138kV	Reconductor line	\$2,401,000	6% (\$144,000)
TES FILER CITY STA - JETPORT 138kV L	Reconductor line	\$1,430,000	3% (\$38,000)
ABBOTSFORD - BRIDGE RIVER 138kV	Reconductor line	\$3,680,000	2% (\$86,000)
PLANT CITY - RED TAIL SUBSTATION 138kV	Reconductor line	\$3,072,000	3% (\$105,000)
ST. MARTINS - WADSWORTH 138kV	Reconductor line	\$1,073,000	5% (\$58,000)
TEMPLETON - RED TAIL 138kV	Reconductor line	\$3,711,000	3% (\$127,000)
TEMPLETON SUBSTATION - BUCKHORN 138kV	Reconductor line	\$2,331,000	3% (\$80,000)
ROSLYN - BUCKHORN 138kV	Reconductor line	\$2,023,000	3% (\$69,000)
HOUSTON - ST LOUIS 138kV	Reconductor line	\$902,000	5% (\$48,000)
DUNN - GLEN HEAD 138kV	Reconductor line	\$1,681,000	3% (\$53,000)
CEDARBURG S - DUNN 138kV	Reconductor line	\$3,814,000	3% (\$121,000)
ST LOUIS - NUCLEAR FUELS 138kV L	Reconductor line	\$2,473,000	6% (\$150,000)
TES FILER CITY STATION - BRIDGE RIV 138kV	Reconductor line	\$1,617,000	2% (\$37,000)
S BELLEVILLE - NICHOLS 138kV	Reconductor line	\$2,749,000	5% (\$133,000)
PLANT CITY - CEDARBURG S 138kV line	Reconductor line	\$4,431,000	3% (\$151,000)
ST LOUIS - NICHOLS 138kV	Reconductor line	\$2,183,000	5% (\$105,000)
BIDGE RIVER - MIDSET 138kV line	Reconductor line	\$2,300,000	3% (\$66,000)
PURDY - CHATOM 138kV L	Reconductor line	\$3,340,000	5% (\$171,000)
	Total allocated to project		\$6,168,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.